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Method for operating a nuclear reactor using calculation of the departure from nucleate boiling ratio

Abstract

A method for operating a nuclear reactor comprises acquiring a plurality of quantities characterizing the operation of the nuclear reactor; and calculating at least one Departure from Nucleate Boiling Ratio using a deep neural network. The entries of the deep neural network are determined by using the acquired quantities. The deep neural network includes at least two hidden layers of at least five neurons each. The method further includes calculating the deviations between the at least one calculated Departure from Nucleate Boiling Ratio and a plurality of predetermined reference threshold values and formulating a control signal for a reactor control system by using the calculated deviations. The control signal is an automatic reactor shutdown or alarm. The method also includes emergency shutdown of the nuclear reactor or emission of an alarm signal if relevant.

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Background/Summary

(1) The present disclosure relates in general to the protection and/or monitoring of the core of a pressurized water nuclear reactor against a boiling crisis.

BACKGROUND

(2) Maintaining the integrity of the first safety barrier (fuel cladding) is a major objective in the design of nuclear reactors, and is the subject of a safety demonstration carried out, among other things, by simulating the behavior of the reactor during families of accidental transients, using numerical calculation codes.

(3) All this analysis is recorded in the final safety report for the facility, which is required by the nuclear safety authority to grant operating authorization for the reactor.

(4) The integrity of the fuel cladding is guaranteed by the absence of fusion in the center of the fuel pellets, and by the absence of a boiling crisis (Departure from Nucleate Boiling, DNB) on the surface of the cladding. These conditions must be met at all points in the core.

(5) The boiling crisis is characterized by a degradation of the thermal exchange between the heat transfer fluid of the primary circuit and the surface of the fuel cladding, which can lead to a caefaction phenomenon and a possible loss of integrity of the first barrier.

(6) Each nuclear reactor is equipped with a protection system, whose role is to guarantee the safety of the reactor for a certain number of accident situations. The protection system typically includes a protection chain known as lowDBNRDeparture from Nuclear Boiling Departure from Nuclear Boiling Ratio DNBR. In reactors with four primary loops such as the 1300 MWe, N4 and EPR, the low DBNRDBNR protection chain ensures the protection of the nuclear reactor against the boiling crisis phenomenon during category 2 accidental transients, characterized by an insertion of reactivity with moderate dynamics (uncontrolled withdrawal of control units, untimely dilution of the primary fluid leading to a decrease in the concentration of boron, untimely cooling of the primary circuit, etc.).

- (7) To this end, the protection system includes an information processing unit (called a digital integrated protection and control system or DIPC) on 1300 MWe and N4 reactors), which integrates a functional unit known as the “low DBNR algorithm”. This algorithm calculates the safety margins online with respect to the boiling crisis. To do so, it calculates the DBNR, and compares the calculated value to a limit value taking into account the reactor design uncertainties and the uncertainties in reconstruction of the DBNR value. The low DBNR protection chain causes an automatic shutdown of the reactor if the reconstructed value is below the limit.
- (8) The Departure from Nuclear Boiling or DBNR is defined as the ratio between the critical heat flux and the local heat flux. The local heat flux corresponds to the heat flux emitted by the fuel rods through the cladding. The critical heat flux is the value of the heat flux leading to damage to the fuel rod cladding, which depends on the local thermo-hydraulic conditions of the fluid.
- (9) The low DBNR algorithm uses a simplified thermo-hydraulic model of the core, adjusted at the design stage by corrective functions (bias curves) acting on the local thermo-hydraulic quantities, in order to ensure a conservative response to the DBNR calculated by the 3D thermo-hydraulic reference code.
- (10) This simplified thermo-hydraulic modeling enables calculation of the local thermal flux from the measured quantities characterizing the power distribution of the reactor and the local thermo-hydraulic conditions allowing the evaluation of the critical thermal flux.
- (11) In addition, the low DBNR algorithm uses a predetermined critical flux correlation to calculate the critical thermal flux. This correlation is determined empirically from experimental data. The design of this correlation shows a complex and costly development. Its multidimensional polynomial-type analytical formula is postulated a priori and manually adjusted in successive steps during the development process in order to meet certain statistical criteria.
- (12) The low DBNR calculation algorithm is not very precise, so that it is necessary to allow large margins to guarantee safety during nuclear reactor operations. This is very penalizing for reactor control. In addition, although simplified, the thermo-hydraulic modeling requires sophisticated numerical processing using microprocessor systems that allow the algorithm to converge with times compatible with protection of the nuclear reactor.

SUMMARY

- (13) In this context, the present disclosure aims to propose a method for operating a nuclear reactor that does not have the above defects.
- (14) To this end, the present disclosure relates to a method for operating a nuclear reactor comprising a core, the method containing the following steps: acquisition of a plurality of quantities characterizing the operation of the nuclear reactor; calculation of at least one Departure from Nuclear Boiling using a deep neural network, the entries of the deep neural network being determined using the acquired quantities, the deep neural network comprising at least two hidden layers of at least five neurons each; calculation of the deviations between the at least one calculated Departure from Nuclear Boiling from a plurality of predetermined reference threshold values; generation of a control signal for a reactor control system, using the calculated deviations, the control signal being selected from a list containing at least the following values: automatic reactor shutdown or alarm; do nothing; emergency shutdown of the nuclear reactor by the control system when the elaborated control signal has the value “automatic reactor shutdown”, or emission of an alarm signal by the control system when the elaborated control signal has the value “alarm”.
- (15) The low DBNR algorithm has the major defect that the local thermo-hydraulic quantities are reconstructed with a simplified physical model: single channel, single phase model, without thermo-hydraulic correlation other than the critical flux correlation (pressure drop, void rate, . . .).
- (16) The use of a deep neural network enables determination of the critical heat flux ratio as a function of the current values of the limiting quantities characterizing the operation of the nuclear reactor, much more precisely and much faster than with the low DBNR algorithm implemented today.

(17) The values predicted by the deep neural network are much closer to the values calculated by the reference thermo-hydraulic code. The reconstruction error of the DBNR is greatly reduced. As a consequence, the operating margins of the nuclear reactor are much better.

(18) The operating method may also have one or more of the following characteristics, considered individually or in any technically possible combination: the said acquired quantities include neutron flux measurements made by neutron flux detectors, the neural network entries containing the axial power distribution in the core of the nuclear reactor, reconstructed from the acquired neutron flux measurements; the axial power distribution is a vector having a plurality of components, each component corresponding to a mean power of the nuclear reactor per axial slice at a given axial position, each component defining one of the entries of the deep neural network; neutron flux detectors are ex-core detectors arranged outside the core; neutron flux detectors are in-core detectors permanently inserted in the core; the nuclear reactor comprises a pressure vessel containing the core and at least one primary loop, the primary loop comprising hot and cold branches fluidly connected directly to the vessel, a pressurizer controlling a pressure in the primary loop, and a primary pump ensuring the circulation of primary fluid in the primary loop, the acquired quantities comprising, for the or each primary loop, one or more of the following quantities: pressurizer pressure, primary fluid flow rate in the primary loop, temperature of the primary fluid in the hot and cold branches; the neural network entries include one or more of the following quantities: pressure of the pressurizer, primary fluid flow rate in the primary loop, temperature of the primary fluid in the cold branch; the neural network entries include the thermal power released by the core; the thermal power released is determined by calculation, using at least the pressurizer pressure, the primary fluid flow rate, the temperatures of the primary fluid in the hot and cold branches, acquired for at least one primary loop; the thermal power released is determined by calculation, using the acquired neutron flux measurements; the entries of the neural network include the enthalpy rise factor; the acquired quantities include a current insertion position of the core reactivity control devices, the enthalpy rise factor being determined by implementing the following operations: calculation of a radial power peak factor using the acquired current insertion position; calculation of the enthalpy rise factor using the calculated radial power peak factor, the reconstructed axial power distribution and the determined thermal power output; the enthalpy rise factor is reconstructed using the axial power distribution; the nuclear reactor comprises multiple primary loops, a Departure from Nuclear Boiling being calculated for each primary loop using the deep neural network using at least said quantities acquired for said primary loop; the detectors are distributed in several sets, the detectors of a same set being staggered vertically above each other, a minimum Departure from Nuclear Boiling being calculated for each set of in-core neutron flux detectors using the deep neural network with the entries determined using at least the neutron flux measurements made by the in-core neutron flux detectors (31) of said set; in the deviation calculation step, a set of deviations is calculated between each calculated Departure from Nuclear Boiling and the plurality of predetermined reference values; in the step of generating a control signal, generating a provisional control signal for each calculated Departure from Nuclear Boiling using the corresponding set of deviations, the provisional control signal being selected from the list comprising at least the following values: automatic reactor shutdown or alarm; do nothing; the control signal being elaborated by using at least several of the provisional control signals; the critical heat flux ratio is calculated by the deep neural network using the following formula:
$$Y(X_{sub.0}) = [\prod_{i=1}^{nH+1} \sigma_{sub.i} \cdot \text{Math.A}_{sub.i}] \cdot \text{Math.X}_{sub.0}$$
 where Y is the calculated Departure from Nuclear Boiling $X_{sub.0}$ is the entries vector; nH is the number of hidden layers of the deep neural network; $A_{sub.i}$ is a predetermined matrix of dimensions $N_{sub.i} \times N_{sub.i-1}$, $N_{sub.i}$, where N_i is the number of neurons in layer i and N_{i-1} is the number of neurons of the layer i-1; $\sigma_{sub.i}$ is a predetermined nonlinear operator; $A_{sub.i}$ is a matrix acting as an affine transformation on the components of the vector $X_{sub.i-1}$, the vector $X_{sub.i-1}$ being the vector whose components are determined by the layer of neurons i-1; $\sigma_{sub.i}$ acts as a sigmoid function

on all the components of the vector A_{i-1} , the vector X_{i-1} being the vector whose components are determined by the neuron layer $i-1$; the method comprises a deep neural network design step, the design step comprising the following sub-steps: constitution of a database comprising at least 100,000 reactor core states, each state being defined by a set of values of said magnitudes characterizing the operation of the nuclear reactor and by a minimum Departure from Nuclear Boiling value calculated by a reference 3D thermal-hydraulic calculation code using the set of values, said states being chosen to cover a plurality of predetermined accident situations of category 2, the reactor control system being provided to handle said accident situations; learning of the deep neural network, using at least part of the database; validation of the deep neural network by using a part of the database independent of that used for the learning phase, and testing of the neural network obtained by coupling with accidental neutronic and thermal-hydraulic codes, making it possible to simulate accidental transients requiring the intervention of the control system. (19) According to a second aspect, the present disclosure relates to a nuclear reactor comprising: a core; a control system; an information processing unit, configured to implement the operating method having the above characteristics.

Description

BRIEF SUMMARY OF THE DRAWINGS

- (1) Other features and advantages will emerge from the detailed description, given below as an indication and in no way limiting, with reference to the appended figures, among which are the following:
- (2) FIG. 1 is a step diagram showing the method of the present disclosure;
- (3) FIG. 2 is a simplified schematic representation of a nuclear reactor intended for the implementation of the method in FIG. 1;
- (4) FIG. 3 is a schematic representation of the steps of acquisition and calculation of the method in FIG. 1, for a first embodiment of the present disclosure;
- (5) FIG. 4 is a schematic representation of the deep neural network used in the method in FIG. 1; and
- (6) FIG. 5 is a schematic representation of the steps of acquisition and calculation of the method in FIG. 1, for a second embodiment of the present disclosure.

DETAILED DESCRIPTION

- (7) The operating method shown schematically in FIG. 1 is intended to be implemented in a nuclear reactor of the type shown in FIG. 2.
- (8) This nuclear reactor is typically a Pressurized Water Reactor (PWR).
- (9) Alternatively, it is intended to be used in any other suitable nuclear reactor.
- (10) The nuclear reactor 1 has a core 3.
- (11) The core 3 typically consists of a plurality of prismatic nuclear fuel assemblies arranged one against the other. Each nuclear fuel assembly comprises a plurality of nuclear fuel rods, held in position relative to each other by grids.
- (12) Each rod has a tubular cladding made of a metal such as zircaloy, with nuclear fuel pellets stacked inside the cladding.
- (13) The nuclear reactor 1 also includes a pressure vessel 5 containing a core 3 and at least one primary loop 7.
- (14) Typically, the nuclear reactor has several primary loops 7, for example three or four primary loops 7.
- (15) The or each primary loop 7 includes hot and cold branches 9, 11 connected fluidically directly to the vessel 5, a pressurizer 13 controlling the pressure in the primary loop 7, and a primary pump 15 providing circulation of primary fluid in the primary loop 7. The pressurizer 13 is generally

common to the different primary loops.

(16) Typically, the primary loop **7** also has a steam generator **17**, located upstream of the primary pump **15**.

(17) The hot branch **9** fluidly connects a primary fluid outlet manifold from the vessel **5** to the primary fluid inlet water box of the steam generator **17**. The pressurizer **13** is connected to hot branch **9**.

(18) An intermediate branch **19** of the primary circuit, often referred to as the “U branch”, connects the primary fluid outlet water box of the steam generator **17** to the suction of the primary pump **15**. The cold branch **11** connects an outlet of the primary pump **15** to a primary fluid inlet of the tank **5**.

(19) The nuclear reactor **1** also has core reactivity control devices **21**.

(20) These control devices **21** are typically designated by the name of “control clusters” and include a plurality of rods made of a neutron-absorbing material.

(21) The nuclear reactor **1** also includes mechanisms, not shown, configured to selectively move the control members **21** inside the vessel **5**, so as to insert or extract them from core **3**.

(22) The nuclear reactor **1** also has a control system **23**.

(23) This control system **23** is typically a protection system, configured to cause an emergency shutdown of the nuclear reactor in the event of an accident situation.

(24) To do this, the control system **23** is configured to send an automatic reactor shutdown command to the drive mechanism of the control bodies **21**. Such an order causes the rapid insertion of the control bodies into the core **3** of the nuclear reactor.

(25) Alternatively, the control system **23** is a monitoring system, configured to generate an alarm when the nuclear reactor approaches its operational limits.

(26) According to another variant, the control system **23** acts as both a monitoring system and a protection system.

(27) The nuclear reactor **1** is also equipped with a plurality of sensors, configured to measure a plurality of quantities characterizing the operation of the nuclear reactor **1**.

(28) Thus, the nuclear reactor **1** has temperature sensors **23**, **25** measuring the temperature of the primary fluid in the hot and cold branches **9**, **11** respectively.

(29) The nuclear reactor **1** includes a sensor **27**, measuring the pressure in the pressurizer **13**.

(30) The nuclear reactor also has a sensor **29** measuring the rotation speed of the primary pump **15**.

(31) The nuclear reactor **1** also has neutron flux detectors **31**.

(32) According to a first embodiment of the present disclosure, these detectors are ex-core detectors arranged outside the core **3**.

(33) The detectors **31** are distributed around the vessel **5**.

(34) Typically, they are multi-stage ex-core chambers, with **6** sections, for example, arranged one above the other along the central axis of the vessel.

(35) The detectors **31** are used to measure the neutron flux emitted by the core **3** at different axial levels.

(36) The nuclear reactor **1** includes a sensor **33** of the position of the core reactivity control devices **21**.

(37) The nuclear reactor also includes an information processing unit **35**, configured to implement an operating method that will be described below.

(38) The information processing unit is, for example, an element of the control system **23**. It typically consists of a processor and a memory related to the processor. Alternatively, the data processing unit is made in the form of programmable logic components, such as FPGAs (Field-Programmable Gate Array), or in the form of dedicated integrated circuits, such as ASICs (Application-Specific Integrated Circuits).

(39) The various sensors **23**, **25**, **27**, **29**, **31**, **33** provide information to the data processing unit **35**.

(40) The method for operating the nuclear reactor **1** according to the first embodiment will now be described, with reference to FIGS. **1**, **3** and **4**.

(41) The method consists of the following steps: step **S10**: acquisition of a plurality of quantities characterizing the operation of the nuclear reactor **1**; step **S12**: calculation of at least one critical heat flux ratio (CHFR) using a deep neural network **37**, shown in FIG. 4, the entries of the deep neural network **37** being determined using the acquired quantities, the deep neural network **37** comprising at least two hidden layers of at least five neurons each; step **S14**: calculation of the deviations between the at least one calculated Departure from Nuclear Boiling (CHFR) and a plurality of predetermined reference values; step **S16**: formulation of a control signal for the control system of the reactor **23**, using the calculated deviations, the control signal being chosen from the list comprising at least the following values: automatic reactor shutdown or alarm; do nothing; step **S18**: emergency shutdown of the reactor by the control system **23**, when the elaborated control signal has the value “automatic reactor shutdown”, or emission of an alarm signal by the control system **23** when the elaborated control signal has the value “alarm”.

(42) The acquired quantities include at least neutron flux measurements made by the neutron flux detectors **31**, allowing reconstruction of the axial power distribution $P(z)$ in core **3**, as described below.

(43) The acquired quantities also include, for the single or each primary loop **7**, one or more of the following quantities: pressure p of the pressurizer, primary fluid flow Q in the primary loop, primary fluid temperature in the hot and cold branches T_{bc} and T_{bf} .

(44) Typically, all of the above quantities are acquired.

(45) The pressurizer pressure p and primary fluid temperatures in the hot and cold branches T_{bc} and T_{bf} are supplied directly from sensors **27**, **23**, **25**.

(46) The primary fluid flow rate Q is reconstructed using the speed of rotation of the primary pumps, measured by the sensor **29**. It is determined by calculation, using for example a relation directly indicating the primary fluid flow rate from the rotation speed.

(47) The acquired quantities also include the current insertion position of the core reactivity control bodies **21**, provided by the sensor **33**.

(48) Thus, as illustrated in FIGS. 1 and 3, the **S10** acquisition step typically includes a sub-step **S20** reading of the value provided by the sensors **23**, **25**, **27**, **29**, **31**, **33**, with certain quantities characterizing the operation of the reactor being read directly from these sensors. These are in particular the pressure p of the pressurizer, and the hot and cold branch temperatures T_{bc} and T_{bf} .

(49) The acquisition step **S10** also includes a sub-step **S22**, during which at least one other quantity characterizing the operation of the nuclear reactor is determined by calculation from the measured values. This typically relates to the reconstruction of the primary fluid flow Q , or the application of filters on the quantities acquired in step **S20**.

(50) The step of calculation of at least one DBNR **S12** comprises a sub-step **S24** of determination of the entries of the deep neural network **37** using the acquired quantities, and a sub-step **S25** of calculation of the at least one DBNR using the deep neural network **37** by using the entries determined in sub-step **S24**.

(51) The entries of the neural network **37** include at least the axial power distribution $P(z)$ in the core **3** of the nuclear reactor, reconstructed from the acquired neutron flux measurements.

(52) The axial power distribution $P(z)$ is a vector having a plurality of components, each component corresponding to a mean power level of the nuclear reactor at a given axial position. Each component defines one of the entries of the deep neural network.

(53) The number of components depends on the degree of accuracy desired for the reconstruction of the power distribution, and thus on the desired accuracy of the DBNR calculation. For example, the axial power distribution has as many components as there are sections in each detector **31**: the signals from the different detector sections define the deep neural network entries.

(54) Alternatively, the axial power distribution has more components than the number of sections in each detector **31**. A module for reconstructing the axial power distribution from the values coming from the different sections of the detectors provides from a dedicated algorithmic processing (such

as a polynomial interpolation) the more finely meshed axial power distribution, each component of which defines one of the deep neural network entries.

(55) In sub-step S24, the power of the nuclear reactor **1** at a given axial level is thus calculated from the measured neutron fluxes, using for example analytical relationships directly giving the power level as a function of the neutron flux and the thermal power released by the Pth core, reconstructed as described below.

(56) The entries of the neural network **37** also include the thermal power released by the Pth core.

(57) In sub-step S24, the heat power output Pth is determined by calculation, using at least the pressurizer pressure p, the primary fluid flow rate Q, and the primary fluid temperatures in the hot and cold branches Tbc and Tbf.

(58) The calculation method is known and will not be detailed here.

(59) Typically, a calculation of the thermal power Pth released for each primary loop **7** is performed, using the quantities acquired for said primary loop **7**.

(60) Entries to the neural network **37** also include the enthalpy elevation factor FAH.

(61) In sub-step S24, the enthalpy rise factor $F\Delta H$ is determined by implementing the following operations: calculation of a radial power peak factor $F_{xy}(z)$, using the current insertion position of the acquired core **21** reactivity control bodies; calculation of the enthalpy rise factor $F\Delta H$ using the calculated radial power peak factor $F_{xy}(z)$, the reconstructed axial power distribution $P(z)$ and the determined thermal power release Pth.

(62) The radial power peak factor $F_{xy}(z)$ and the enthalpy rise factor $F\Delta H$ are calculated using known formulas, which will not be detailed here.

(63) Preferably, an enthalpy rise factor calculation $F\Delta H$ is performed for each primary loop **7**, using the released thermal power Pth calculated for said primary loop **7**.

(64) When the nuclear reactor **1** comprises multiple primary loops **7**, a Departure from Nuclear Boiling DBNR is preferably calculated in sub-step S25 for each primary loop **7**, using the deep neural network **37**. For this purpose, the deep neural network **37** uses the entries determined with at least the acquired quantities for the corresponding primary loop **7**.

(65) Alternatively, a single DBNR Departure from Nuclear Boiling is calculated using the deep neural network **37**. It is calculated, for example, using entries determined by any processing of the acquired quantities for the primary loops, adapted for the DBNR evaluation.

(66) As can be seen in FIG. 4, the deep neural network **37** comprises an input layer **39** located on the left of FIG. 4, an output layer **41** located on the right of FIG. 4, and a plurality of intermediate layers **43**, also called hidden layers. The number nH of intermediate layers **43** is greater than or equal to two, preferably greater than or equal to five, and preferably greater than or equal to ten.

(67) This makes it possible to account for a large number of different situations in the nuclear reactor.

(68) Each hidden layer **43** comprises at least five neurons **45**, preferably at least seven neurons **45**, and preferably at least ten neurons **45**.

(69) The input layer **39** has as many input neurons as there are entries.

(70) For example, input layer **39** has one input neuron for each of the entries p, Q, Tbf, Pth, $F\Delta H$. It also has one input neuron for each of the components of the axial power distribution $P(z)$.

(71) For example, input layer **39** has eleven input neurons, if the axial power distribution $P(z)$ is a six-component vector.

(72) The output layer **41** has a single output neuron, corresponding to the minimum value of the PCR calculated for the corresponding primary loop.

(73) Advantageously, the entries are subjected to a first post-processing before being introduced into the neural network **37** in step S22 of FIG. 1.

(74) The first post-processing consists, for example, in applying dynamic compensation terms, typically advance-delay filters, allowing each acquired quantity to be the image of the corresponding physical parameter. This post-processing compensates for the instrumentation

response time (cold and hot branch temperature measurements of the vessel), the dropping time of the core reactivity control devices, and the response time of the processes and processing algorithms. The functional nature of information processing by the neural network **37** allows the need to use filters to be considerably reduced, compared with conventional algorithms.

(75) The entries are preferably the subject of a second post-processing before being introduced into the neural network, by normalization of each of the quantities or components.

(76) The entries constitute a vector, called an input vector and noted X_0 . Each hidden layer i of the neural network elaborates a vector of values X_i , by processing the vector of values X_{i-1} from the immediately preceding layer $i-1$.

(77) Thus, for each hidden layer i , i being between 1 and n_H ,

$X_{i,j} = \sigma_{i,j} \cdot \sum_{k=1}^{N_{i-1}} A_{i,j,k} \cdot X_{i-1,k}$ where $A_{i,j,k}$ is a predetermined matrix of dimensions $N_{i,j} \times N_{i-1}$, $N_{i,j}$, where N_i is the number of neurons in layer i and N_{i-1} is the number of neurons in layer $i-1$; σ_i is a predetermined non-linear operator.

(78) In other words, the Departure from Nuclear Boiling is calculated by the deep neural network **37** using the following formula:

$Y(X_0) = [\prod_{i=1}^{n_H+1} \sigma_{i,j} \cdot \sum_{k=1}^{N_{i-1}} A_{i,j,k} \cdot X_{i-1,k}]$ where Y is the calculated Departure from Nuclear Boiling.

(79) Each matrix $A_{i,j,k}$ acts as an affine transformation on the components of the vector X_{i-1} . In other words:

$\sum_{k=1}^{N_{i-1}} A_{i,j,k} \cdot X_{i-1,k} = \sum_{k=1}^{N_{i-1}} W_{i,j,k} \cdot X_{i-1,k} + b_{i,j}$ where $W_{i,j,k}$ is the matrix of synaptic weights that link the N_i neurons of layer i to the N_{i-1} neurons of layer $i-1$, and b_i is the vector of the N_i dimensional biases of layer i .

(80) For the last layer we note that the matrix A_{n_H+1} is of dimension $1 \times N_{n_H}$ and that $\sigma_{n_H+1} = 1$: there is no application of a non-linear transfer function to establish the Y signal of the last layer.

(81) The non-linear operators σ_i each act as a non-linear function, for example sigmoid, on all the components of the $A_i X_{i-1}$ vector. A sigmoid function is a function expressed as follows:

(82) $f(x) = \frac{1}{1 + e^{-lx}}$ where l is a specially parameterized constant.

(83) In step **S14**, for each primary loop, a set of deviations is calculated between the critical heat flux ratio calculated for said primary loop and the plurality of predetermined reference values.

(84) Thus, as many deviation calculations are performed as there are primary loops.

(85) These values are operating reference thresholds, called protection (or monitoring) thresholds, which incorporate the uncertainties of the method, that is, the measurement uncertainties and the calculation uncertainties.

(86) Typically, the calculated deviations are displayed alphanumerically or graphically on at least one screen for real-time monitoring of the protection and operating margins of the unit for each primary loop.

(87) When a single DBNR is calculated, a single set of deviations is calculated in step **S14**.

(88) Step **S16** is used to interpret the sets of deviations found in step **S14**.

(89) To do this, a provisional control signal is elaborated for each primary loop, using the set of deviations calculated for that loop.

(90) When the control system is a protection system, the provisional control signal is chosen from the list containing at least the following values: automatic reactor shutdown, do nothing.

(91) When the control system is a monitoring system, the provisional control signal is chosen from the list containing at least the following values: alarm, do nothing.

(92) When the control system is a protection and monitoring system, the temporary control signal is chosen from the list containing at least the following values: automatic reactor shutdown, alarm, do nothing.

(93) The control signal sent to the control system of reactor **23** is elaborated using the provisional control signals of all primary loops.

(94) After having interpreted the deviations noted in step S14, step S16 consists of applying voting logics for the signals elaborated for each loop of the reactor. For example, if the majority of the provisional control signals have the value “automatic reactor shutdown”, then the control signal sent to control system 23 is “automatic reactor shutdown”.

(95) In step S18, the control system 23 receives the control signal developed in step 16, and acts accordingly. For example, it triggers an automatic shutdown of the reactor 1 by insertion of the reactivity control devices of the core 21 if the control signal is “automatic reactor shutdown”, or the emission of an alarm signal if the control signal elaborated has the value “alarm”.

(96) Preferably, the method also includes a step S26 for the design of the deep neural network 37. The design step S26 consists of the following sub-steps: S28: creation of a database containing at least 100,000 reactor states, S30: learning of the deep neural network, using at least part of the database.

(97) Advantageously, the database includes at least 500,000 reactor states, preferably at least 1,000,000 reactor states.

(98) Each state is defined by a set of values of the quantities characterizing the operation of the nuclear reactor, and by a Departure from Nuclear Boiling value. The said Departure from Nuclear Boiling value is calculated by a 3D thermo-hydraulic reference code using the corresponding set of values of the quantities characterizing the operation of the nuclear reactor.

(99) The 3D core thermal-hydraulic calculation code is a high-precision code, for example the FLICA code. Typically, Monte-Carlo sampling techniques are used to generate the database.

(100) The states are chosen to cover a plurality of predetermined accident situations leading to trigger the intervention of the reactor's generic low-level DBNR protection chain.

(101) Predetermined accident situations are, for example, those referred to as Category 2 in the nuclear reactor safety report, which require the intervention of the generic low-level DBNR protection chain.

(102) The ranges of variation of the quantities characterizing the operation of the nuclear reactor are also chosen in such a way that they cover the range of validity of the critical thermal flux correlation.

(103) The data are generated in such a way that they optimize the detection of physical symmetry during the learning phase by the neural network.

(104) Preferably, the input variables are normalized.

(105) The learning sub-step S30 is performed on only a part of the database created in sub-step S28. For example, it uses between 50% and 90% of the database, typically 80%.

(106) The learning phase consists of the following operations: i) Determination of the optimal neuronal structure by carrying out a sensitivity study dedicated to the choice of model hyperparameters, this step may require the use of optimization algorithms (of the genetic algorithm type): number of hidden layers, number of neurons per layer, choice of the transfer function imposed by the phenomenology of the boiling crisis phenomenon (e.g. derivable and having to present a strong non-linear character due to the analytical expression of the critical flux correlation); ii) Development of a learning algorithm optimized specifically for the generated database, allowing acceleration and stabilization of the convergence process, and to improve convergence towards more robust solutions by avoiding local minima. This improves the generalization capabilities of the neural network and thus its ability to respond positively to the validation and testing phases described below.

(107) This is achieved, for example, by developing an adaptive batch stochastic gradient descent algorithm. Updating of the synaptic weight and bias tensor learning at each cycle is performed on data batches of evolving size by scanning the entire learning base, parameterized by the cycle number.

(108) In addition, inertia methods on the adaptation step of the learning algorithm are advantageously used, in order to improve the efficiency and robustness of the learning phase.

(109) Preferably, step **S26** also includes a sub-step **S32** of validation of the deep neural network determined in sub-step **S30**. This validation phase is performed on a second part of the database, which was not used in sub-step **S30**. Typically, the entire part of the database that was not used in step **S30** is used in step **S32**. This enables validation of the deep neural network predictions developed at the end of step **S30** by comparison with the predictions generated with the reference code, and to characterize the generalizability of the predictive model thus developed. Sensitivity studies are preferably carried out during sub-step **S32**. If the validation phase is unsatisfactory, we return to sub-step **S30**, then modify the neuronal structure (that is, the number of hidden layers, number of neurons per layer), or the hyperparameters of the learning algorithm, in order to restart the learning step aimed at determining the matrices of the synaptic weights and bias vectors of the neural network thus defining the previously defined **Ai** matrices.

(110) Preferably, step **S26** also includes a sub-step **S34** of phenomenological validation (testing) of the deep neural network on database-independent data determined in sub-step **S28**.

(111) During sub-step **S34**, the deep neural network is implemented in coupling with calculation codes simulating accidental transients. These accidental transients are typically those defined in the nuclear reactor safety report.

(112) If this final test phase is not satisfactory, sub-step **S30** is repeated.

(113) Thus, the method described above is particularly adapted to be implemented by a reactor of the type described above.

(114) Conversely, the nuclear reactor **1** described above is particularly adapted to implement the method just described.

(115) In particular, as shown in FIG. 2, the information processing unit **35** comprises: a module **47** for the acquisition of a plurality of quantities characterizing the operation of the nuclear reactor; a module **49** for calculating at least one Departure from Nuclear Boiling using the deep neural network **37**, configured to determine the entries of the deep neural network **37** using the quantities acquired and determined, the deep neural network **37** comprising at least two hidden layers of at least five neurons each; a module **51** for calculating the deviations between the at least one calculated Departure from Nuclear Boiling and a plurality of predetermined reference values; a module **53** for generating a control signal for the reactor control system using the calculated deviations, the control signal being selected from the list comprising at least the following values: automatic reactor shutdown or alarm; do nothing.

(116) The reactor control system **23** is configured to cause an automatic shutdown of the reactor when the elaborated control signal has the “automatic reactor shutdown” value. The reactor control system **23** is configured to generate an alarm when the processed control signal has the value “alarm”.

(117) In other words, the module **47** is configured to implement step **S10** of the method described above. The module **49** is configured to carry out step **S12** of the method described above.

(118) The module **51** is configured to perform step **S14** of the method described above and the module **53** is configured to perform step **S16** of the method described above.

(119) A second embodiment of the present disclosure will now be described, with reference to FIG. 5. Only the points by which the second embodiment differs from the first will be detailed below. Identical elements or elements performing the same functions will be designated by the same references.

(120) In the second embodiment, the neutron flux detectors **31** are not ex-core detectors arranged outside the core **3**. The neutron flux **31** detectors are in-core detectors, permanently inserted in the core **3**.

(121) They are distributed in several sets, the detectors of the same set being staggered vertically above each other.

(122) The detectors are colons, for example, vertically arranged on rods in the core. Each electron rod allows the neutron flux to be measured at several axial heights in the core, such as six axial

heights for a vertical arrangement of **6** electrons. The nuclear reactor is typically equipped with **12** rods of **6** electrons each, distributed in the core.

(123) The thermal power P_{th} released is determined by calculation, using the acquired neutron flux measurements. This calculation no longer uses the pressurizer pressure p , the primary fluid flow rate Q , the temperatures of the primary fluid in the hot and cold branches T_{bc} , T_{bf} . The calculation formulas used are known and will not be detailed here.

(124) The enthalpy rise factor $F\Delta H$ is reconstructed using the axial power distribution $P(z)$. It is not necessary to reconstruct the factor $F_{xy}(z)$ from the measurement provided by the sensor **33** of the position of the reactivity control bodies of the core **21**.

(125) In step **S12**, a Departure from Nuclear Boiling DBNR is calculated for each set of **31** in-core neutron flux detectors, using the deep neural network **37**, with the entries determined using the neutron flux measurements made by said 31 in-core neutron flux detectors.

(126) From the neutron flux measurements at several axial heights provided by the detector set, a set of parameters $P(z)$, P_{th} and $F\Delta H$, specific for this detector set, is determined. These parameters are used as entries for the neural network **37**.

(127) Other entries are also used, such as pressurizer pressure p , primary fluid flow Q , primary fluid temperature in the cold branch T_{bc} , T_{bf} . These input quantities are the same for all sensor sets.

(128) In the deviation calculation step **S14**, a set of deviations is calculated between each previously determined critical heat flux ratio and the plurality of predetermined reference values.

(129) In the control signal generation step **S16**, a provisional control signal is generated for each calculated critical heat flux ratio using the corresponding set of deviations.

(130) The control signal is elaborated using at least several of the provisional control signals according to the voting logic described above.

(131) A third embodiment of the present disclosure will now be briefly described. Only the points by which the third embodiment differs from the second will be detailed below. Identical elements or elements performing the same functions will be designated by the same references.

(132) In the third embodiment, the data processing unit **35** has a continuous 3-dimensional mapping of the power in the reactor core.

(133) This mapping is one of the quantities characterizing the operation of the reactor. The other quantities, for example T_{bf} , p and Q , are acquired as in the first and second modes.

(134) Some entries of the deep neural network **37** are determined by computation, using the 3D mapping of power distribution in the core, thus replacing the quantities $P(z)$, P_{th} and $F\Delta H$ previously described for the first embodiment of the present disclosure.

(135) Advantageously, the components of the 3D power distribution constitute the entries of a dedicated neural structure, called convolution layer, which enables optimization of the extraction of characteristics of the power distribution in order to elaborate the deep neural network, and in particular to optimize its structure for a hardware implementation. The output of this convolution layer defines part of the entries of the deep neural network **37**. In particular, this output replaces the vector of the components of the axial power distribution at the input of the deep neural network **37**, as well as the size $F\Delta H$. This neural structure, adapted to process spatially structured data, replaces matrix multiplication by a mathematical convolution operation between the input power distribution of the structure and a series of kernels (or filters) whose degrees of freedom are adjusted during the training of step **S30**. The convolution layer also includes a non-linear processing of the convolution outputs through non-linear transfer functions, as well as a third step allowing to introduce invariants under local geometric transformations (such as translations). These steps can be repeated in the elaboration of the optimal neuronal structure.

(136) The method for the exploitation of the present disclosure may present multiple variants.

(137) The quantities characterizing the operation of the nuclear reactor, acquired in step **S10**, and the entries of the deep neural network may not correspond exactly to the list described above. Other

quantities may be acquired. Some quantities may not be acquired. Some entries may not be used. Other entries may be added.

(138) In any case, the power distribution has an important impact on the location in the core and the value of the minimum margin of the DBNR.

(139) Taking this into account improves the accuracy of the DBNR reconstruction. The quantities characterizing the operation of the nuclear reactor are not necessarily acquired in the way described above and shown in FIGS. 2, 3 and 5. Some quantities may not be read directly by the sensors equipping the nuclear reactor. They could be derived from other measured quantities or values from the operating system of the nuclear reactor.

(140) The convolution layer described in the third embodiment could also be implemented in the first two realization modes in order to process the axial power distribution $P(z)$ in the neuronal structure. The other entries of the neural network being unchanged.

(141) The method described above has multiple advantages.

(142) The response time is extremely short, for example in the millisecond range. This is obtained in particular because the method does not require the implementation of convergence loops, only the parameters specific to the deep neural network being necessary, the unit calculation operations are simple and easily parallelizable if the hardware solution allows it.

(143) Consequently, it frees up computing capacity for other parts of the nuclear reactor control and instrumentation system.

(144) This performance is obtained by the fact that the complexity of the underlying physics is encoded by the parameterization of the deep neural network from the database. This allows the deep neural network to reconstruct the DBNR value of the thermo-hydraulic reference code, by simply using the synaptic weights and transfer functions of the neural structure.

(145) The DBNR computation provided by the deep neural network is very reliable, and the response is close to the one provided by the 3D thermo-hydraulic reference code (here FLICA). The conservatism of the response is adjustable.

(146) The quality of the response provided by the deep neural network is notably due to the fact that no reconstruction of intermediate local thermo-hydraulic variables is performed without fine physical modeling. The reliability of the response provided is a function of the fineness of the mesh of the database used for learning the neural network. This only costs computing time, before commissioning in the nuclear reactor. The method is also more robust to the propagation of uncertainties on the input data (initiated by random fluctuations in the measurements from the core instrumentation) which improves the validation and hardware qualification stage.

Claims

1. A method for operating a nuclear reactor comprising a core, the method comprising the following steps: acquiring a plurality of quantities characterizing the operation of the nuclear reactor; calculating at least one departure from nucleate boiling ratio using a deep neural network, entries of the deep neural network being determined using the acquired quantities, the deep neural network comprising at least two hidden layers of at least five neurons each; calculating deviations between the at least one calculated departure from nucleate boiling ratio and a plurality of predetermined reference threshold values; generating a control signal for a control system of the reactor, using the calculated deviations, the control signal being selected from a list comprising at least the following values: automatic reactor shutdown and alarm; and either: emergency shutting down the nuclear reactor by the control system based on the generated control signal having the value of automatic reactor shutdown, or emitting an alarm signal by the control system based on the generated control signal having the value of alarm; wherein the acquired quantities comprise neutron flux measurements made by neutron flux detectors, and the entries of the neural network comprise an axial power distribution in the core of the nuclear reactor, wherein the axial power

distribution is reconstructed from the acquired neutron flux measurements.

2. The method according to claim 1, wherein the axial power distribution is a vector having a plurality of components, each component corresponding to an average power of the nuclear reactor per axial slice at a given axial position, each component defining one of the entries of the deep neural network.
3. The method according to claim 1, wherein the neutron flux detectors are ex-core detectors arranged outside the core.
4. The method according to claim 1, wherein the neutron flux detectors are in-core detectors permanently inserted in the core.
5. The method according to claim 4, wherein the entries of the neural network include the thermal power released by the core and wherein the thermal power released is determined by calculation, using the acquired neutron flux measurements.
6. The method according to claim 4, wherein the detectors are distributed in several sets, the detectors of a same set being vertically staggered one above the other, a minimum departure from nucleate boiling ratio being calculated for each set of in-core neutron flux detectors with the aid of the deep neural network with the entries determined using at least the neutron flux measurements made by the in-core neutron flux detectors of the set.
7. The method according to claim 6, wherein in the deviation calculation step, a set of deviations is calculated between each calculated departure from nucleate boiling ratio and the plurality of predetermined reference values; and wherein in the step of generating the control signal, elaborating a provisional control signal for each calculated departure from nucleate boiling ratio using the corresponding set of deviations, the provisional control signal being selected from the list comprising at least the following values: automatic reactor shutdown and alarm; the control signal being elaborated using at least several of the provisional control signals.
8. The method according to claim 5, wherein in the deviation calculation step, a set of deviations is calculated between each calculated departure from nucleate boiling ratio and the plurality of predetermined reference values; and wherein in the step of generating the control signal, elaborating a provisional control signal for each calculated departure from nucleate boiling ratio using the corresponding set of deviations, the provisional control signal being selected from the list comprising at least the following values: automatic reactor shutdown and alarm; the control signal being elaborated using at least several of the provisional control signals.
9. The method according to claim 1, wherein the nuclear reactor comprises a pressure vessel containing the core and at least one primary loop, the primary loop comprising hot and cold branches fluidly connected directly to the vessel, a pressurizer controlling a pressure in the primary loop, and a primary pump ensuring the circulation of primary fluid in the primary loop, the acquired quantities comprising, for the at least one primary loop, one or more of the following quantities: pressure of the pressurizer, flow rate of primary fluid in the primary loop, and temperature of the primary fluid in the hot and cold branches.
10. The method according to claim 9, wherein the neural network entries have one or more of the following variables: pressurizer pressure, flow rate of primary fluid in the primary loop and temperature of the primary fluid in the cold branch.
11. The method according to claim 9, wherein the entries of the neural network include a thermal power released by the core and wherein the thermal power released is determined by a calculation using, acquired for the at least one primary loop, at least the pressurizer pressure, the primary fluid flow rate and the temperatures of the primary fluid in the hot and cold branches.
12. The method according to claim 9, wherein the nuclear reactor comprises several primary loops, a departure from nucleate boiling ratio being calculated for each primary loop with the aid of the deep neural network using at least the acquired quantities for the primary loop.
13. The method according to claim 1, wherein the entries of the neural network include the thermal power released by the core.

14. The method according to claim 1, wherein the entries of the neural network include the enthalpy rise factor.

15. The method according to claim 14, wherein the neutron flux detectors are ex-core detectors arranged outside the core and the entries of the neural network include the thermal power released by the core, wherein the thermal power released is determined by a calculation using, acquired for the at least one primary loop, at least the pressurizer pressure, the primary fluid flow rate and the temperatures of the primary fluid in the hot and cold branches, wherein the acquired quantities include a current insertion position of core reactivity control bodies, the enthalpy rise factor being determined by carrying out the following operations: calculating a radial power peak factor using the current acquired insertion position; and calculating the enthalpy rise factor using the calculated radial power peak factor, the reconstructed axial power distribution and the determined thermal power output.

16. The method according to claim 14, wherein the neutron flux detectors are in-core detectors permanently inserted in the core and wherein the enthalpy rise factor is reconstructed using the axial power distribution.

17. The method according to claim 1, wherein the departure from nucleate boiling ratio is calculated by the deep neural network using the following formula:

$$Y(X_{sub.0}) = [\Pi_{sub.i=1}^{sup.i=nH+1} \sigma_{sub.i}(\text{Math.A}_{sub.i} \cdot \text{Math.X}_{sub.0})]$$
 where Y is the calculated departure from nucleate boiling ratio; $X_{sub.0}$ the being a vector having components, said components being said entries of the deep neural network determined using the acquired quantities; nH is the number of hidden layers of the deep neural network; A_i is a predetermined matrix of dimensions $N_{sub.i} \times N_{sub.i-1}$, where $N_{sub.i}$ is the number of neurons in layer i and $N_{sub.i-1}$ is the number of neurons in layer i-1; σ_i is a predetermined non-linear operator.

18. The method according to claim 17, wherein each hidden layer i of the neural network elaborates a vector X_i , by processing the vector X_{i-1} issued from the immediately preceding layer i-1, according to the following formula:

$$X_{sub.i} = \sigma_i(\text{Math.A}_i \cdot \text{Math.X}_{i-1})$$
 wherein each matrix A_i acts as an affine transformation on the vector X_{i-1} .

19. The method according to claim 17, wherein each hidden layer i of the neural network elaborates a vector X_i , by processing the vector X_{i-1} from the immediately preceding layer i-1, according to the following formula:

$$X_i = \sigma_i(\text{Math.A}_i \cdot \text{Math.X}_{i-1})$$
 wherein each non-linear operator σ_i acts as a sigmoid function on vector $A_i X_{i-1}$.

20. The method according to claim 1, wherein the method comprises a deep neural network design step, the design step containing the following sub-steps: constituting a database containing at least 100,000 reactor core states, each state being defined by a set of values of the quantities characterizing the operation of the nuclear reactor and by a minimum departure from nucleate boiling ratio value calculated by a reference 3D thermo-hydraulic calculation code using the set of values, the states being chosen to cover a plurality of predetermined category 2 accident situations, the control system of the reactor being provided to handle the accident situations; learning of the deep neural network, using at least part of the database; validating the deep neural network using a part of the database independent of that used for the learning, and testing of the neural network obtained by coupling with accidental neutronic and thermal-hydraulic codes making it possible to simulate accidental transients requiring the intervention of the control system.
